

More formulas are given than is necessary. Use the right one based on your needs.

For Ungroup or Raw data: Population mean, $\mu = \frac{\sum_{i=1}^N X_i}{N}$ Sample mean, $\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$ For Odd n, Median = $\left(\frac{n+1}{2}\right)^{th}$ value For Even n, Median = $\frac{1}{2}\left[\left(\frac{n}{2}\right)^{th} \text{ value} + \left(\frac{n}{2} + 1\right)^{th} \text{ value}\right]$ Mode = The value which has occurred most in the data	For Group data: Population mean, $\mu = \frac{\sum_{i=1}^K f_i X_i}{N}$ Sample mean, $\bar{x} = \frac{\sum_{i=1}^k f_i x_i}{n}$ OR Sample mean, $\bar{x} = \frac{\sum_{i=1}^n f_i x_i}{\sum_{i=1}^n f_i}$ Empirical Relationship: Mean - Mode = 3 (Mean - Median)	Median = $L_m + \frac{\frac{n}{2} - F}{f_m} \times h$ Mode = $L_o + \frac{\Delta_1}{\Delta_1 + \Delta_2} \times h$ OR, Median = $L_o + \frac{\frac{n}{2} - F_{-1}}{f_m} \times h$ Mode = $L_o + \frac{(f_o - f_{-1})}{(f_o - f_{-1}) + (f_o - f_1)} \times h$
Pearson's coefficient of skewnes = $\frac{(\text{mean} - \text{mode})}{\text{Standard Deviation}}$ Pearson's coefficient of skewnes = $\frac{3(\text{mean} - \text{median})}{\text{Standard Deviation}}$ Bowley's coefficient of skewness = $\frac{(Q_3 - Q_2) - (Q_2 - Q_1)}{Q_3 - Q_1}$ Interquartile Range, $IQR = Q_3 - Q_1$ Lower fence = $Q_1 - 1.5 * IQR$ Upper fence = $Q_3 + 1.5 * IQR$	Formulas for finding percentiles/quartiles/deciles: For raw data, $P_i = \frac{in}{100}$ for i = 1, 2, 3, ..., 99 $Q_i = \frac{in}{4}$ for i= 1, 2, 3 $D_i = \frac{in}{10}$ for i = 1, 2, 3, ..., 9	
For Ungroup or Raw data: Population Mean Deviation = $\frac{\sum X_i - \mu }{N}$ Sample Mean Deviation = $\frac{\sum x_i - \bar{x} }{n}$ Population variance, $\sigma^2 = \text{Var}(X) = \frac{\sum_{i=1}^N (X_i - \mu)^2}{N}$ Sample variance, $s^2 = \text{var}(X) = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}$ OR, Sample variance, $s^2 = \frac{\sum x_i^2 - n\bar{x}^2}{n-1}$ Population CV, $C_v = \frac{\sigma}{\mu} * 100\% = \text{Value in \%}$ Correlation coefficient, r $= \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$ $= \frac{\sum x_i y_i - n\bar{x}\bar{y}}{\sqrt{\{\sum x_i^2 - n\bar{x}^2\} \{\sum y_i^2 - n\bar{y}^2\}}}$ Simple Linear Regression Model: $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$ $\hat{\beta}_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} = \frac{\sum x_i y_i - n\bar{x}\bar{y}}{\sum x_i^2 - n\bar{x}^2}$ $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$	For Group data: Population Mean Deviation = $\frac{\sum f_i X_i - \mu }{N}$ Sample Mean Deviation = $\frac{\sum f_i x_i - \bar{x} }{n}$ Population variance, $\sigma^2 = \text{Var}(X) = \frac{\sum_{i=1}^K f_i (X_i - \mu)^2}{N}$ Sample variance, $s^2 = \text{var}(X) = \frac{\sum_{i=1}^k f_i (x_i - \bar{x})^2}{n-1} = \frac{\sum_{i=1}^k f_i x_i^2 - n\bar{x}^2}{n-1}$ Population SD, $\sigma = \text{SD}(X) = \sqrt{\text{Var}(X)}$ Sample SD, $s = \text{sd}(X) = \sqrt{\text{var}(X)}$ Sample CV, $c_v = \frac{s}{\bar{x}} * 100\% = \text{Value in \%}$ Quartile, $Q_i = L_o + \frac{\frac{i \times n}{4} - F_{-1}}{f} \times h$; for i = 1, 2, 3 Decile, $D_i = L_o + \frac{\frac{i \times n}{10} - F_{-1}}{f} \times h$; for i = 1, 2, ..., 9 Percentile, $P_i = L_o + \frac{\frac{i \times n}{100} - F_{-1}}{f} \times h$; for i = 1, 2, ..., 99	